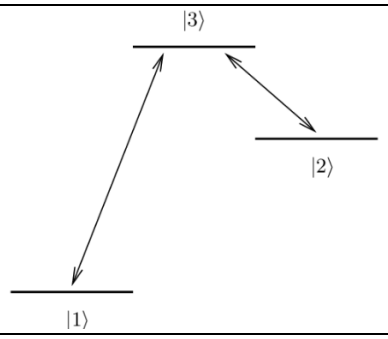
Methanol (CH_3OH)

Levels Data									
Level	J	K_a	K_c	Sym	v_t	g	Energy		
							cm^{-1}	K	
3	5	1	5	A+	0	44	34.0983	49.05991	
2	6	0	6	A+	0	44	33.8759	48.73987	
1	4	0	4	A+	0	44	16.1335	23.21251	

Transitions Data			
Transition	Frequency (MHz)	A_{ij}	d_{ij} (D)
$3 \rightarrow 1$	538570.5530	9.864×10^{-4}	0.73652
$3 \rightarrow 2$	6668.5640	1.593×10^{-9}	0.67936

We now describe the relationship between the two-level and three-level population densities:

$$N_{123}^0 = N_1^0 + N_{23}^0 \quad , \quad r = \frac{N_1^0}{N_{23}^0}$$
$$= (r+1) N_{23}^0$$

Where N_{123}^0 is the number of atoms initially in all three states, and N_{23}^0 is the number of atoms initially in only two states. If we assume that $N_3^0 = aN_1^0$ and $N_2^0 = bN_3^0$ so:

$$r = \frac{N_1^0}{N_{23}^0} = \frac{1}{a(1+b)}$$

In a two-level system, the relationship between parameters b and θ_0 is:

$$\cos\theta_0 = \frac{N_3^0 - N_2^0}{N_3^0 + N_2^0} = \frac{1-b}{1+b} \;\Rightarrow\; b = \frac{1-\cos\theta_0}{1+\cos\theta_0}$$

The time scale is given by:

$$T_0 = \frac{2\epsilon_0 \hbar}{n_0 L |d_{32}|^2 k_{32}}$$

Thus, the relationship between the two-level and three-level time scales is given by:

$$\frac{T_{123}^0}{T_{23}^0} = \frac{N_{23}^0}{N_{123}^0} = \frac{1}{1+r}$$

Use S255 Data

S255IR-NIRS3, $v_{lsr} = 6.93$ km s^{-1}	
$\lambda = 0.044987$ (m)	$T_1 = 1.64 \times 10^{7}$ (s)
$\gamma = 1.56 \times 10^{-9}$ (s^{-1})	$T_2 = 1.55 \times 10^{6}$ (s)
$n_0 = 1.17 \times 10^{-6}$ (m^{-3})	$\Lambda_0 = 7.15 \times 10^{-14}$ (m^{-3}s^{-1})
$L = 2.10 \times 10^{13}$ (m)	$\Lambda_1 = 2.60 \times 10^{-13}$ (m^{-3}s^{-1})
$\theta_0 = 4.14 \times 10^{-10}$ (rad)	$T_t = 8.10 \times 10^{6}$ (s)

If we want to reproduce the two-level plots, we must set:

1. $N_1^0 \ll N_{23}^0 \;\Rightarrow\; r \to 0$ or $a \to \infty$
2. $d = 0$

$$T_0 = \frac{2\epsilon_0 \hbar}{n_0 L |d_{32}|^2 k_{32}} = \frac{8\pi}{3\gamma n_0 L \lambda^2} \simeq 108031 \text{ (s)}$$

$$r = 0 \;\Rightarrow\; \begin{cases} n_{123}^0 = n_{23}^0 = n_0 \\ T_{123}^0 = T_{23}^0 = T_0 \end{cases}$$

$T_1 \simeq 152\ T_{123}^0$

$T_2 \simeq 14\ T_{123}^0$

$\Lambda_1 = 2.4 \times 10^{-2} \left(\frac{n_{123}^0}{T_{123}^0} \right)$

$T_t \simeq 75\ T_{123}^0$